MEDXAI CLINICAL MRI RESEARCH REPORT

Report ID:	62997593-191e-43ff-aa2d-fe2576b00905
Date/Time:	2026-08-13 07:10:41.804207
Scan Filename:	test.jpeg
Diagnosis Classification:	**Non Demented**
Model Confidence:	97.65%

AI Prediction Summary

The MEDXAI EfficientNetB0 model classified the submitted MRI image as **Non Demented** with a model confidence of **97.65%**. The model produced a high-confidence classification for the submitted MRI image.

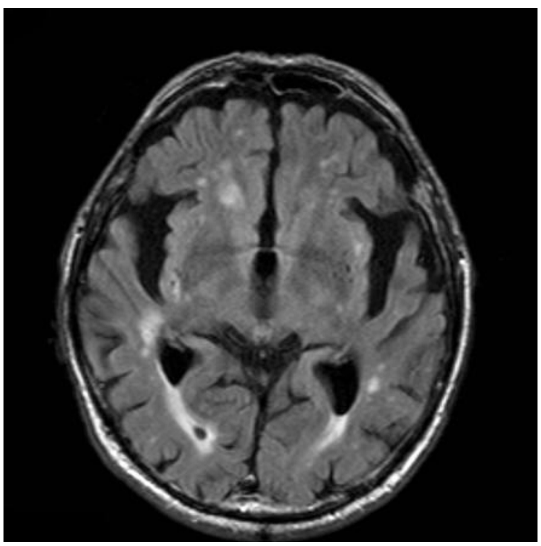
Probability Analysis

The highest model probability is associated with **Non Demented** at **97.65%**. The next highest probability is **Moderate Demented** at **2.33%**, giving a probability difference of **95.32 percentage points**. This indicates a comparatively strong separation between the leading class and the next most likely class.

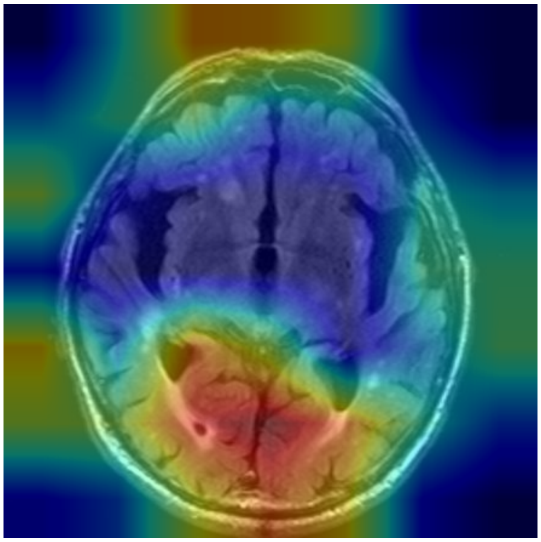
Class Probability Breakdown

Class	Probability
Non Demented	97.65%
Moderate Demented	2.33%
Mild Demented	0.01%
Very Mild Demented	0.00%

Original MRI Scan

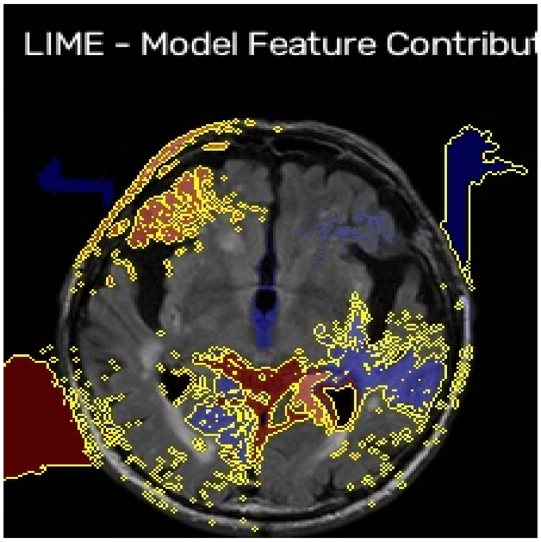


Grad-CAM Explainability Heatmap



The Grad-CAM visualization provides a model-focused representation of the image regions that contributed most strongly to the **Non Demented** classification. Highlighted regions should be interpreted as areas receiving greater influence from the neural network rather than as definitive evidence of disease.

LIME Local Explanation



The LIME visualization provides a local explanation of the model decision by identifying image regions that contributed to the prediction for this individual MRI. These regions describe the behavior of the AI model and should not be interpreted as a direct anatomical diagnosis.

Overall AI Findings

The overall MEDXAI analysis classified the MRI as **Non Demented** with a confidence of **97.65%**. Both Grad-CAM and LIME explainability outputs were successfully generated. The probability distribution describes how the model allocated prediction likelihood across the available classes, while Grad-CAM and LIME provide complementary views of the model's decision process. These explainability methods describe model behavior and do not independently establish the presence, absence, severity, or stage of Alzheimer's disease.

• Leading classification: **Non Demented**

• Leading probability: **97.65%**

• Reported model confidence: **97.65%**

• Explainability interpretation: highlighted regions represent areas that influenced the AI model's decision and should not be treated as independently diagnostic.

MEDICAL DISCLAIMER: *This AI-generated report is produced by the MEDXAI EfficientNetB0 Engine for research and clinical decision-support purposes. The prediction, probability values, Grad-CAM visualization, and LIME explanation describe the behavior of an artificial intelligence model and are not a medical diagnosis. This report must be reviewed by a qualified radiologist, neurologist, or other appropriately qualified healthcare professional before any clinical diagnosis or treatment decision is made.*